

BUS TICKETING SYSTEM

Complete Project Documentation & Presentation Report

PROJECT OVERVIEW

Abstract

A terminal-based Bus Ticketing System built in C language.

Users can book, search, view and cancel tickets.

Data is stored persistently using file handling techniques.

Objectives & Problem

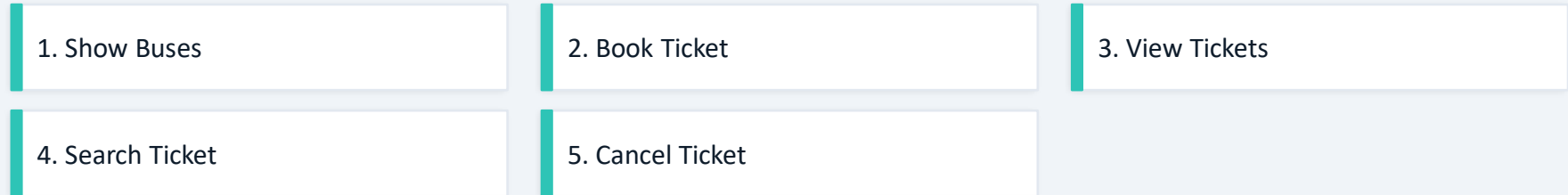
- Learn structures, functions, loops
- Apply switch-case and arrays
- Master file handling in C
- Build a practical management system

Problem: Manual ticket management is time-consuming and error-prone. This project automates the process.

SYSTEM WORKFLOW



Menu Options:



DATA STRUCTURE & FUNCTIONS

```
struct Ticket  
  
    struct Ticket {  
        int ticketID;  
        char passengerName[50];  
        int age;  
        char busName[50];  
        int seatNumber;  
    };
```

Functions Used	
showBuses()	Display bus list
bookTicket()	Book and save ticket
viewTickets()	Display all tickets
searchTicket()	Search by ticket ID
cancelTicket()	Delete a ticket
main()	Main menu controller

FEATURES: INPUT / OUTPUT

Feature 1 — Show Buses

```
● ● ●  
$ ./bus_ticket  
===== BUS TICKET SYSTEM =====  
Enter Choice: 1  
  
AVAILABLE BUSES  
1. Dhaka Express  
2. Green Line  
3. Hanif Enterprise
```

Feature 2 — Book Ticket

```
● ● ●  
Enter Choice: 2  
Passenger Name: Abu  
Age: 22  
Bus Name: Green Line  
Seat Number: 12  
  
Ticket Booked Successfully!  
Ticket ID: 1000
```

Feature 3 — View Tickets

```
● ● ●  


| ID   | NAME  | AGE | BUS              | SEAT |
|------|-------|-----|------------------|------|
| 1000 | Abu   | 22  | Green Line       | 12   |
| 1001 | Rahim | 25  | Hanif Enterprise | 07   |
| 1002 | Karim | 30  | Dhaka Express    | 15   |


```

FEATURES: SEARCH, CANCEL & FILE HANDLING

Feature 4 — Search Ticket



Enter Ticket ID: 1000

TICKET FOUND

Passenger : Abu

Bus : Green Line

Seat : 12

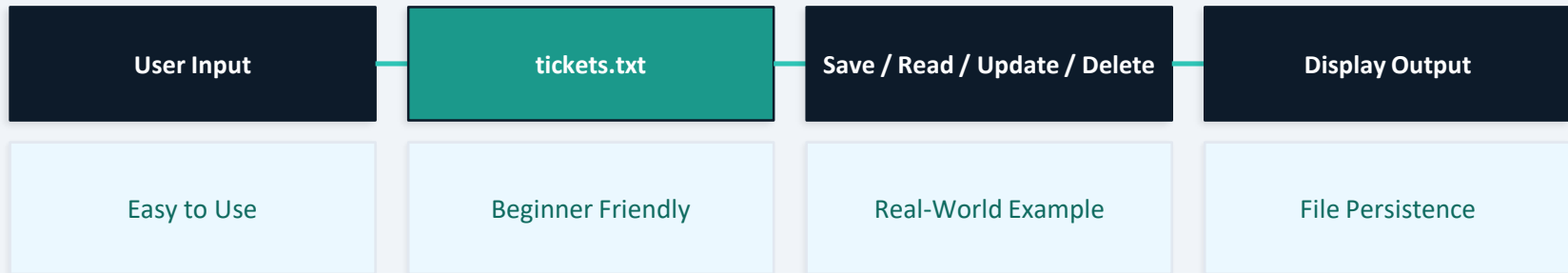
Feature 5 — Cancel Ticket



Enter Ticket ID: 1000

Ticket Cancelled Successfully

File Handling Process



TESTING RESULTS

Test Case	Result
Book Ticket	✓ Pass
View Tickets	✓ Pass
Search Ticket	✓ Pass
Cancel Ticket	✓ Pass

All 4 test cases passed successfully

FUTURE IMPROVEMENTS & CONCLUSION

Admin Login

Secure admin access for system management

Fare Calculation

Automatic pricing based on route & class

Seat Map

Visual interactive seat selection layout

Database Integration

Replace file handling with SQL/SQLite

Bus Schedule Mgmt

Timetable management for all routes

GUI Version

Graphical interface for better UX

Conclusion

The Bus Ticketing System successfully demonstrates structures, functions, loops, switch-case statements and file handling in C programming — making it an excellent academic project showcasing real-world software concepts.